The quadratic correlation matrix need not have full nonnegative rank

A counterexample to NR-03 and an explicit factorization for every size

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Abstract

For subsets $a, b \subseteq [n]$, let $C_n(a,b) = (1 - |a \cap b|)^2$, with every entry prescribed. We construct nonnegative rational factors proving

$$\operatorname{rank}_+(C_n) \leq 2^{n-1} + n + \binom{n}{2} + \binom{n}{4}.$$

At $n = 7$ this gives $\operatorname{rank}_+(C_7) \leq 127 < 128$, disproving the universal full-rank conjecture in NR-03. In fact the displayed bound is strictly below 2^n for every $n \geq 7$. The construction separates the matrix into a component invariant under complementing its row index and three nonnegative correction families. A polynomial identity proves the factorization exactly. The package also supplies all factors for $n = 7$ and construction-independent integer checks of all 16,384 entries. No numerical optimization, freely chosen completion, or assumption about real factor coordinates is used.

1 The precise question and its negative answer

For $M \in \mathbb{R}_{\geq 0}^{m \times q}$, its nonnegative rank is

$$\operatorname{rank}_+(M) = \min\{r : M = WH,\ W \in \mathbb{R}_{\geq 0}^{m \times r},\ H \in \mathbb{R}_{\geq 0}^{r \times q}\}.$$

The repository asks whether $\operatorname{rank}_+(C_n) = 2^n$ for every integer $n \geq 3$, where

$$C_n(a,b) = (1 - |a \cap b|)^2, \qquad a, b \subseteq [n]. \tag{1}$$

In particular, entries with intersection size greater than one are not free parameters [1]. The same conjecture appears as Conjecture 4 in Section 6.4 of Vandaele et al. [2], and is retained in the recent benchmark discussion of Baeckelant et al. [3].

Theorem 1. *For every integer $n \geq 1$, with $\binom{n}{k} = 0$ when $k > n$,*

$$\operatorname{rank}_+(C_n) \leq \min\left\{2^n,\ 2^{n-1} + n + \binom{n}{2} + \binom{n}{4}\right\}. \tag{2}$$

Consequently,

$$\boxed{\operatorname{rank}_+(C_7) \leq 64 + 7 + 21 + 35 = 127 < 128,}$$

and $\operatorname{rank}_+(C_n) < 2^n$ *for every $n \geq 7$.*

This is a complete negative answer to the universally quantified question. It does not assert that the exact nonnegative rank of C_7 is 127, or that 7 is the smallest counterexample dimension.

2 The explicit factors

Let $a^c = [n] \setminus a$. For each column index b, write

$$p = |b|, \qquad d_b = \max\{1, (p-1)^2\}, \qquad D = \operatorname{diag}(d_b). \tag{3}$$

Thus D is a positive diagonal matrix, including at singleton columns. Choose one representative from each complementary pair by taking

$$\mathcal{A} = \{s \subseteq [n] : n \notin s\}, \qquad |\mathcal{A}| = 2^{n-1}.$$

For $T \in \binom{[n]}{4}$, set

$$h_T(a) = \max\{|a \cap T| - 2, 0\}. \tag{4}$$

This function takes values $0, 0, 0, 1, 2$ as $|a \cap T|$ ranges from 0 to 4.

Construct W and V using the following disjoint families of atom indices. The actual right factor will be $H = VD^{-1}$.

Complementary pairs: 2^{n-1} atoms

For each $s \in \mathcal{A}$, use

$$W_{a,s} = \mathbf{1}_{\{a=s \text{ or } a=s^c\}}, \qquad V_{s,b} = (1 - |s \cap b|)^2 (1 - |s^c \cap b|)^2. \tag{5}$$

Singleton columns: n atoms

For each $i \in [n]$, use

$$W_{a,i} = \mathbf{1}_{\{i \notin a\}}, \qquad V_{i,b} = \mathbf{1}_{\{b=\{i\}\}}. \tag{6}$$

Pairs: $\binom{n}{2}$ atoms

For each $S \in \binom{[n]}{2}$, use

$$W_{a,S} = \mathbf{1}_{\{S \subseteq a\}}, \qquad V_{S,b} = 4(p-2)\mathbf{1}_{\{S \subseteq b\}}. \tag{7}$$

Four-element sets: $\binom{n}{4}$ atoms

For each $T \in \binom{[n]}{4}$, use

$$W_{a,T} = h_T(a), \qquad V_{T,b} = 12\mathbf{1}_{\{T \subseteq b\}}. \tag{8}$$

Every entry of W and V is a nonnegative integer. In (7), the indicator can be nonzero only if $p \geq 2$, so the coefficient $p - 2$ cannot give a negative entry. The total number of atoms is

$$r(n) = 2^{n-1} + n + \binom{n}{2} + \binom{n}{4}. \tag{9}$$

We now prove the exact identity $WV = C_n D$. Positive column scaling then gives the desired nonnegative rational factorization $C_n = W(VD^{-1})$.

3 Entrywise proof of the factorization

Fix $a, b \subseteq [n]$, and put $t = |a \cap b|$ and $p = |b|$. Then $|a^c \cap b| = p - t$.

Exactly one complementary pair contains a. Therefore the sum of its family of atom contributions is

$$G = (t-1)^2(p-t-1)^2. \tag{10}$$

The singleton family contributes

$$E = \mathbf{1}_{\{p=1\}}(1-t). \tag{11}$$

There are $\binom{t}{2}$ pairs contained in both a and b, so the pair family contributes

$$P = 4(p-2)\binom{t}{2}. \tag{12}$$

For a four-element set $T \subseteq b$, the row function $h_T(a)$ is one when exactly three elements of T lie in a, two when all four do, and zero otherwise. Counting these two cases gives

$$Q = 12\left((p-t)\binom{t}{3} + 2\binom{t}{4}\right). \tag{13}$$

Lemma 1. *For all real p, t, the following polynomial identity holds:*

$$\begin{aligned}
&(p-1)^2(t-1)^2 - (t-1)^2(p-t-1)^2 \\
&\quad = t(t-1)^2(2p-2-t) \\
&\quad = 2(p-2)t(t-1) + 2(p-t)t(t-1)(t-2) \\
&\qquad\qquad\qquad + t(t-1)(t-2)(t-3).
\end{aligned} \tag{14}$$

Proof. The first equality follows by taking the difference of the squares $(p-1)^2$ and $(p-t-1)^2$. For the second, factor out $t(t-1)$ and use

$$(t-1)(2p-2-t) = 2(p-2) + 2(p-t)(t-2) + (t-2)(t-3).$$

Expanding either side verifies this last equality. □

Proof of the factorization and the upper bound in Theorem 1. If $p = 0$, then $t = 0$, $d_b = 1$, and $(G, E, P, Q) = (1, 0, 0, 0)$. If $p = 1$, then $t \in \{0, 1\}$, $d_b = 1$, $G = P = Q = 0$, and $E = 1 - t = (1-t)^2$. Thus $G + E + P + Q = d_b(1-t)^2$ in these two cases.

If $p \geq 2$, then $d_b = (p-1)^2$ and $E = 0$. Replacing the falling factorials in (14) by binomial coefficients gives

$$d_b(t-1)^2 - G = 4(p-2)\binom{t}{2} + 12(p-t)\binom{t}{3} + 24\binom{t}{4} = P + Q.$$

Thus in all cases $(WV)_{a,b} = d_b(1 - |a \cap b|)^2$. This proves $WV = C_n D$, hence $C_n = W(VD^{-1})$ with $r(n)$ nonnegative rational atoms. The trivial factorization $C_n = I_{2^n} C_n$ gives the other upper bound in (2). □

4 Strictness for every $n \geq 7$

Write $q_n = n + \binom{n}{2} + \binom{n}{4}$. Then $q_7 = 63 < 64$. Pascal's identities imply

$$q_{n+1} - 2q_n = 1 - \binom{n}{2} + \binom{n}{3} - \binom{n}{4}. \tag{15}$$

For $n \geq 7$, we have $\binom{n}{4} = \frac{n-3}{4}\binom{n}{3} \geq \binom{n}{3}$, so the right-hand side of (15) is strictly negative. Induction therefore yields $q_n < 2^{n-1}$ for every $n \geq 7$, proving

$$\operatorname{rank}_+(C_n) \leq 2^{n-1} + q_n < 2^n \qquad (n \geq 7).$$

This completes Theorem 1. Some values of the constructed bound are:

n	Complementary pairs	Singletons	Pairs	Four-sets	Total $r(n)$
7	64	7	21	35	127
8	128	8	28	70	234
9	256	9	36	126	427
10	512	10	45	210	777

Each total is strictly smaller than 2^n. Moreover,

$$\frac{\operatorname{rank}_+(C_n)}{2^n} \leq \frac{1}{2} + \frac{n + \binom{n}{2} + \binom{n}{4}}{2^n} = \frac{1}{2} + O(n^4 2^{-n}).$$

This is an upper estimate, not a determination of the actual limiting ratio.

5 A transparent numerical slice of the certificate

For $n = 7$ and $b = [7]$, we have $p = 7$ and $d_b = 36$. The table gives the integer-scaled contributions for each intersection size t. The singleton correction is zero on this column.

t	Complementary-pair part G	Pair part P	Four-set part Q	$(G + P + Q)/36$
0	36	0	0	1
1	0	0	0	0
2	16	20	0	1
3	36	60	48	4
4	36	120	168	9
5	16	200	360	16
6	0	300	600	25
7	36	420	840	36

The last column is exactly $(1 - t)^2$, including every prescribed value for $t > 1$. All the other columns of C_7 are checked in the same exact way by the complete certificate verifier.

6 Certificate files and reproducibility

The supplied matrices $W \in \mathbb{Z}_{\geq 0}^{128\times 127}$, $V \in \mathbb{Z}_{\geq 0}^{127\times 128}$, and $d \in \mathbb{Z}_{>0}^{128}$ satisfy

$$WV = C_7\,\mathrm{diag}(d), \qquad H = V\,\mathrm{diag}(d)^{-1}, \qquad C_7 = WH.$$

The denominator values are $1, 4, 9, 16, 25, 36$. Every entry of W is in $\{0, 1, 2\}$, and every entry of V is at most 36.

Indexing and data

Rows and columns follow integer masks $0, \ldots, 127$; bit i denotes element $i + 1$. Atoms are ordered as 64 complementary pairs (representative masks $0, \ldots, 63$), seven singletons, 21 pairs, and 35 four-sets. Pairs and four-sets follow lexicographic coordinate-tuple order; labels are included.

Integer matrices, denominators, and atom labels are in `data/factors_n7.json`. CSV matrices are `W_n7.csv` and `H_scaled_n7.csv`; denominators are in `column_denominators_n7.csv`. The additional `H_n7_rational.csv` stores H as exact fraction strings: its product with W equals C_7 itself.

Exact checks

Python 3.10 or later suffices, without third-party packages, a compiler, network access, or an optimization solver:

```
python3 code/verify_certificate.py
python3 verify_all.py
```

The first command reads the certificate without importing its generator. It checks dimensions, nonnegativity, positive denominators, and all 16,384 entries by integer multiplication, including 5,103 zeros and 11,281 positive entries.

The full suite also regenerates the certificate, checks the polynomial identity coefficient by coefficient, verifies all 349,524 entries across $n = 1, \ldots, 9$, multiplies the unscaled rational CSV factors, and rejects six damaged certificates. Successful logs are included. An optional, separate C++ implementation checks the integer CSV files; it passed with both GCC and Clang.

7 Scope, prior work, and review status

The previous package proving $\mathrm{rank}_+(C_4) = 16$ is preserved unchanged under `provenance/`, with a successful rerun log. The present construction does not depend on it and is compatible with it: $r(4) = 19$, $r(5) = 36$, and $r(6) = 68$ give no nontrivial upper bounds at those sizes.

The $n = 7$ counterexample fully settles the universal question negatively without determining exact ranks at $n = 5, 6, 7$. It does not determine the extension complexity of the entire correlation polytope: factors for this submatrix need not factor the full slack matrix.

The submitted proof and checks were developed with ChatGPT assistance. On 13 September 2026, a separate Codex AI agent independently audited the entire argument and exact target, returning PASS for the complete negative resolution; its report and independently written exact checker accompany this submission. Under the repository's evidence rules [4], this supports *Solved*. This is informal AI-agent review, not external human peer review or formal verification. No Lean verification was performed.

A A small construction-independent verifier

The following is sufficient to check the supplied counterexample. It uses no knowledge of the four atom families: its only mathematical target is (1). The full verifier adds stronger input diagnostics and summary statistics.

```
import json
from pathlib import Path

c = json.loads(Path("data/factors_n7.json").read_text())
W, V, d = c["W"], c["H_scaled"], c["denominators"]

def check(ok, message):
    if not ok:
        raise ValueError(message)

check(c["n"] == 7 and c["r"] == 127, "Wrong metadata")
check(len(W) == 128 and all(len(x) == 127 for x in W),
      "Wrong left-factor dimensions")
check(len(V) == 127 and all(len(x) == 128 for x in V),
      "Wrong right-factor dimensions")
check(len(d) == 128 and all(type(x) is int and x > 0 for x in d),
      "Invalid denominators")
check(all(type(x) is int and x >= 0 for row in W + V for x in row),
      "Invalid factor entry")
for a in range(128):
    for b in range(128):
        actual = sum(W[a][k] * V[k][b] for k in range(127))
        target = d[b] * (1 - (a & b).bit_count()) ** 2
        check(actual == target, f"Bad entry: {a}, {b}")
print("PASS: rank_+(C_7) <= 127 < 128")
```

References

[1] A. Townsend, *Open Problems in Numerical Linear Algebra*, entry NR-03, "Full nonnegative rank of the quadratic correlation matrix." Canonical statement inspected in this conversation; the historical page records Last checked September 11, 2026.
https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/nonnegative-and-positive-factorizations/NR-03

[2] A. Vandaele, N. Gillis, F. Glineur, and D. Tuyttens, *Heuristics for Exact Nonnegative Matrix Factorization*, arXiv:1411.7245, Section 6.4, Conjecture 4.
https://arxiv.org/html/1411.7245

[3] T. Baeckelant, A. Vandaele, and N. Gillis, *Computing Lower Bounds on the Nonnegative Rank via Non-Convex Optimization Solvers*, arXiv:2605.14058v2, July 6, 2026, Section 6.7 and Appendix A.4.
https://arxiv.org/html/2605.14058v2

[4] *Open Problems in Numerical Linear Algebra*, README, "Problem status." The distinction between a complete manuscript, an independently audited resolution, and formal verification.
https://github.com/ajt60gaibb/OpenProblemsInNLA#problem-status